Proc. of International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA 2025)
7-9 August 2025, Antalya-Türkiye

2025 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA) | 979-8-3315-3562-9/25/$31.00 ©2025 IEEE | DOI: 10.1109/ACDSA65407.2025.11166338

Customer Complaint Classification with Large Language Models

Merve Güllü
R&D Department
Türk Telekom
Ankara, Türkiye
merve.gullu@turktelekom.com.tr

Ercan Atagün
Department of Computer Engineering
Düzce University
Düzce, Türkiye
ercanatagun@duzce.edu.tr

Serdar Biroğul
Department of Computer Engineering
Düzce University
Düzce, Türkiye
serdarbirogul@duzce.edu.tr,
serdarbirogul@ndu.edu.az

Necaattin Barışçı
Department of Computer Engineering
Gazi University
Ankara, Türkiye
nbarisci@gazi.edu.tr

***Abstract*—Automatic classification of customer complaints is of great importance for companies to detect customer problems early, optimize service processes and increase customer satisfaction. Customer complaint data provided as textual data consists of very insightful information in terms of addressing the problems encountered. In this study, deep learning-based approaches (LSTM, CNN), transformer-based models (BERT, RoBERTa) and large language models (GPT-2, Llama3.1-8B) are compared using two customer complaint datasets with different class distributions. The results show that LLMs achieve higher accuracy and F1 scores in both training and test phases. In particular, the fine-tuning process applied to the large language models enabled them to be tailored to the problem and contributed to the higher performance of LLMs compared to other models.**

***Keywords*—Customer complaint, LLM, GPT, Llama, Bert, RoBERTa**

I. Introduction

With the rapid increase in digitalization, customer experience has become a key factor for businesses to gain a competitive advantage. Especially in the telecommunications sector, where intense competition and rapid technological changes are experienced, ensuring and sustaining customer satisfaction is a strategic imperative [1]. In this context, practical analysis of customer complaints makes it possible to address individual dissatisfaction, identify systematic service problems, and develop improvement strategies [2].

In telecom, customers often communicate their complaints in free text format. These texts contain multi-layered information such as emotions, experiences, technical problems, and expectations. However, this richness brings with it significant challenges. Factors such as text data's inherent unstructured nature, context-dependent language variability, semantic complexity, and polysemy reveal the limitations of traditional text processing techniques [3]. Therefore, the effectiveness of natural language processing (NLP) methods is directly related to the capacity of the models to understand and interpret the context.

In recent years, Large Language Models (LLMs) have revolutionized the field of NLP, enabling more accurate, context-aware analysis of texts [4]. These transformer-based models can be trained on large datasets to understand word relations, context links, and semantic patterns better. Thanks to these capabilities, LLMs have shown significant success in complex tasks such as customer complaint classification [5], sentiment analysis [6], summarization [7], and cause-effect inference [8].

In this study, customer complaint texts from the telecommunications industry will be analyzed and automatically classified using an LLM-based approach. The goal is to categorize customer complaints quickly, accurately, and scalably so that service providers can take data-driven actions to improve customer satisfaction. Despite the growing interest in using LLMs in sectoral text analysis, studies on telecom complaints are limited. In this context, the proposed approach aims to make academic and industry contributions by targeting industry-specific problem areas.

II. Related Works

Research on consumer feedback analysis has consistently been significant in academic and industrial contexts. To enhance customer satisfaction and facilitate complaint analysis, these methodologies are also crucial for telecommunications [9], [10], transportation [11], banking applications, reservation systems such as hotels [12], automotive enterprises [13], and various companies utilizing e-commerce platforms. The early methodologies in this domain predominantly concentrated on

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text categorization and sentiment analysis tasks employing conventional machine learning methods. Techniques including Naive Bayes, Support Vector Machines (SVM), and Decision Trees have been used alongside basic text representations such as TF-IDF, word2vec, and n-gram to classify complaints as positive or negative or to categorize them into distinct classifications [9]–[14]. Nevertheless, these techniques inadequately capture contextual linkages within the text and exhibit notably poor performance in polysemous or structurally complicated complaint texts.

The adoption of deep learning methodologies has grown progressively to address these restrictions. Long Short-Term Memory (LSTM) networks are distinguished by their capacity to learn long-term dependencies in sequential data, producing effective outcomes in identifying customer complaints and analyzing sentiment in their remarks [15], [16]. Nonetheless, the emergence of pre-trained language models in the domain of NLP has led to models such as Bidirectional Encoder Representations from Transformers (BERT) and Generative Pre-trained Transformer (GPT) achieving far greater success in contextual comprehension [17]. These models have begun to be commonly employed in text analysis within highly customer-interactive industries such as telecoms [18].

In the past three years, there has been a notable rise in utilizing LLMs for analyzing customer complaints [19]. Models created by OpenAI, including GPT-3 and GPT-4, along with systems like Google's T5 and PaLM, have been effectively utilized across multiple sectors for tasks such as classifying customer complaints, automatic summarization, sentiment extraction, and generating suggestions [20]. In telecommunications research, the application of these models is still exemplified by a restricted number of instances. Gewaly et al. [10] advocate for the automated categorization of customer complaints utilizing traditional machine learning algorithms. In contrast, Wu et al. [21] underscore the efficacy of BERT in multilingual complaint analysis. Li et al. [22] conducted a comparative investigation of LLMs for predicting the Customer Service Evaluation Metric in the telecoms industry. Their analysis demonstrates that open-source LLMs performance surpasses closed-access LLMs. Nevertheless, most existing studies either focus on a restricted range of complaint types utilizing traditional machine learning techniques or fail to assess the capacity of LLMs to comprehend context thoroughly and adequately.

This study's primary contribution is creating a scalable system capable of automatically and accurately classifying customer complaints using LLM-based methodologies on a broad and diverse array of complaint texts pertinent to the telecommunications industry. The objective is to derive insights into the fundamental reasons for customer complaints by conducting explanatory studies of the model's results. This approach addresses the sectoral deficiencies in the existing LLM-based complaint analysis literature while enhancing firms' operational decision-making processes.

III. Dataset and Preprocessing

In this study, two different customer complaint datasets are used to evaluate the performance of LLM and deep learning based approaches. This diversity was chosen in order to comparatively analyze the classification performance of the models against different data structures and distributions.

The first dataset contains 5,629 complaints about Comcast's customer service posted on online platforms [23]. Each complaint is accompanied by a rating value defined by the users. The dataset was created to qualitatively and quantitatively analyze Comcast's chronic customer service problems. The unbalanced distribution of rating classes in this dataset is particularly concentrated in the first class (high complaint rate), and the effects of this imbalance on classification performance were also analyzed (see Figure 1).

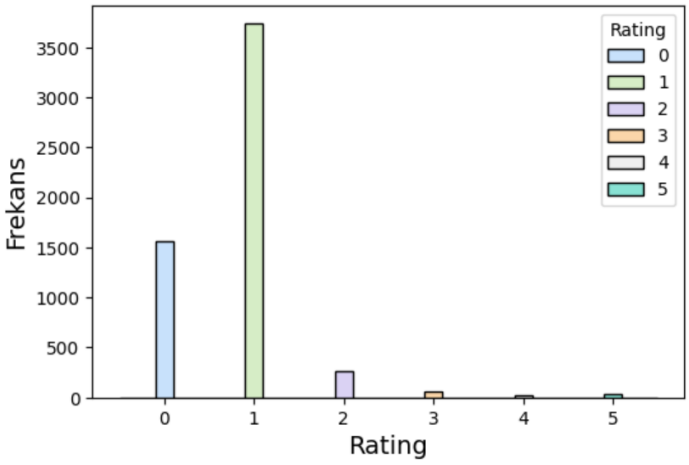


Fig. 1. Dataset Rating Distribution

The second dataset, Customer Complaints Dataset [24], contains a total of 162,421 complaint records made by consumers about various products and services. Each record contains the text of the complaint and the relevant product category. Product categories include financial services such as "Credit Reporting", "Debt Collection", "Mortgages and Loans", "Credit Cards" and "Retail Banking". This dataset was used to test the overall classification capabilities of the models (see Figure 2), as it contains a wider and more diverse set of classes.

The main purpose of using two datasets together is to comparatively demonstrate how the models perform both in a small and focused complaint structure (Comcast dataset) and in a large and class-diverse data environment (Customer Complaints Dataset).

Various text preprocessing steps were applied to both datasets to improve classification performance. These steps include lowercase conversion, removal of punctuation marks, numbers, special characters, stop words and lemmatization. After these steps, the GloVe (Global Vectors for Word Representation) preprocessed word embedding method was used for vector representation of the text data. Glove is a widely

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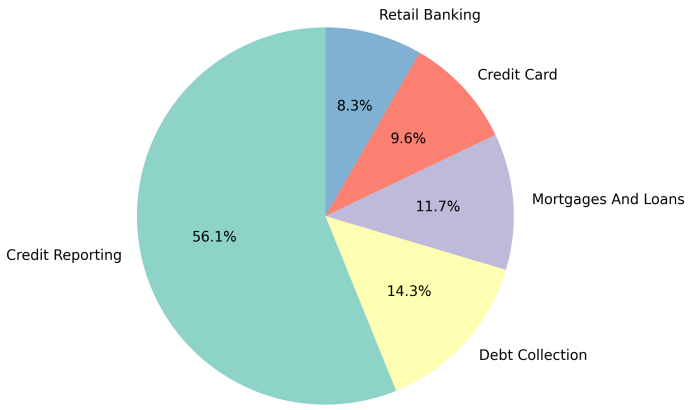


Fig. 2. Dataset Label Distribution

used method for vectorizing words in the text classification problem. Glove also performs well on imbalanced datasets [25]. Glove has different dimensional versions such as 50d, 100d, 200d, and 300d vectors. In this study, the 100d Glove vector, which is frequently preferred in the literature, is used.

IV. Method

This study includes experimental studies on two datasets with six different methods: llama3.1, gpt2, RoBERTa, BERT, LSTM and CNN. All methods were used for 80% training and 20% test in these two datasets.

A. Convolutional Neural Network

Convolutional Neural Network (CNN) is a neural network architecture that performs feature extraction using convolution to learn spatial and local relationships in the input data. CNN is used in text classification problems by applying convolution on text data digitized by word embedding methods [26].

In this study, 5-layer CNN is applied, and the model parameters and details are given in Table I.

TABLE I.
Parameters of CNN

hyperparams	values
Embedding vector	Glove Embedding
Number of hidden layers	5
Time window size	5
Number of neurons	128
Number of epochs	15
Activation function	relu
Optimizer	adam
Learning Rate	0.001

B. Long Short-Term Memory

Long Short-Term Memory (LSTM) is a special type of Recurrent Neural Networks that solves the vanishing gradient problem, which is the fundamental problem of Recurrent Neural Network (RNN) architecture. LSTM has been widely used in text classification problems thanks to its ability to capture long dependencies on sequential data [27].

In this study, 5-layer LSTM is applied, and the model parameters and details are given in Table II.

TABLE II. Parameters of LSTM

hyperparams	values
Embedding vector	Glove Embedding
Number of hidden layers	3
Time window size	100
Number of neurons	128
Number of epochs	15
Activation function	tanh
Optimizer	adam
Learning Rate	0.001

C. Bidirectional Encoder Representations from Transformer

Bidirectional Encoder Representations from Transformers (BERT) is a language model developed by Google that has revolutionized the NLP field. Based on the Transformer architecture, BERT makes sense of context bidirectionally by processing texts both left-to-right and right-to-left simultaneously. Its ability to capture bidirectional semantic context has enabled better feature extraction from textual expressions and provided significant performance improvements in the text classification problem [28].

In this study, the pre-trained "bert-base-uncased" model was used, and the parameters and details of the training and test process are given in Table III.

TABLE III.
Parameters of BERT

hyperparams	values
base model	bert-base-uncased
Embedding Vector	bert-base-uncased
epochs	3
max sequence length	128
return tensors	pt
optimizer	AdamW
batch size	16
Learning Rate	0.00002

D. Robustly Optimized BERT Approach

Robustly Optimized BERT Approach (RoBERTa) is an optimized BERT architecture model developed by Facebook AI Research. While maintaining the basic principles of BERT, it has achieved superior performance in language understanding and text classification tasks by systematically improving parameters such as training data size, batch size, training time, and maximum token length. Unlike BERT, it uses masked language modeling (MLM) instead of next-sentence prediction (NSP) and is made more efficient by dynamic masking. RoBERTa, both the improvements in the pre-training phase and its performance during the fine-tuning phase have led to superior performance on text classification problems in [29]. It offers higher performance and generalization than BERT.

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In this study, the pre-trained "roberta-base" model was used and the parameters and details of the train and test process are given in Table IV.

TABLE IV.
PARAMETERS OF ROBERTA

hyperparams	values
base model	roberta-base
Embedding Vector	roberta-base
epochs	3
max sequence length	100
return tensors	tf
optimizer	AdamW
batch size	16
Learning Rate	0.00005

E. Large Language Model Meta AI - LLaMa

Llama 3.1 is an 8 billion parameter language model developed by Meta. As a large language model with a powerful transformer-based architecture, Llama 3.1 can be used effectively in text classification problems [30]. Llama3.1 is available in 8B, 70B, and 405B versions. Llama3.1 8B is used in this study. The parameter details used in the training and test process with Llama3.1:8b in this study are given in Table V.

F. Generative Pre-trained Transformer - GPT

GPT-2 Large is a transformer-based language model with 774 million parameters developed by OpenAI. GPT-2 is especially preferred in short text classification problems thanks to its transformer-based structure, parameter size, and accessibility to run on on-prem systems [31].

This study gives the parameter details used in the training and test process with GPT2 and Llama3.1:8b LLM in Table V.

TABLE V.
PARAMETERS OF LLMS

params	values
Number of epochs	3
Max token lenght	512
Batch size	8
Learning rate	5e-5
Optimizer	AdamW
Loss Function	CrossEntropyLoss
average	weighted
return tensors	pt

V. EXPERIMENTAL RESULTS AND COMPARISONS

In this study, six different methods/models were used to classify two different datasets. As a result of the experimental studies, the success of the models in terms of accuracy, precision, recall and F1-score are compared in detail. The results of the models on the training and test dataset are shown in Table VI,VII,VIII,IX.

The results of the experimental studies using the dataset 1 are presented in Table VI for the training data and Table VII for the test data. When the results are analyzed, the GPT-2 model showed the highest performance among all models with 85.07% accuracy and 83.07% F1 score in the training process and 80.04% accuracy and 77.99% F1 score in the test process. The Llama3.1-8B model achieved an F1 score of 68.61% in the test process, making it the second most successful model after GPT-2.

TABLE VI.
ACCURACY VALUES OF TRAIN OF MODELS DATASET 1

Model	Accuracy	Precision	Recall	F1-Score
Llama3.1-8B	0.7625	0.7139	0.7625	0.7362
gpt2	0.8507	0.8301	0.8507	0.8307
RoBERTa	0.6598	0.4353	0.6598	0.5245
BERT	0.7765	0.3132	0.2694	0.2715
LSTM	0.7601	0.7299	0.7601	0.7065
CNN	0.6638	0.5960	0.6638	0.5353

TABLE VII.
ACCURACY VALUES OF TEST MODELS DATASET 1

Model	Accuracy	Precision	Recall	F1-Score
Llama3.1-8B	0.7323	0.6863	0.7323	0.6861
gpt2	0.8004	0.7716	0.8004	0.7799
RoBERTa	0.6598	0.4354	0.6598	0.5246
BERT	0.7765	0.3133	0.2695	0.2715
LSTM	0.6396	0.5432	0.6396	0.5657
CNN	0.6617	0.7118	0.6617	0.5288

The RoBERTa model, despite being the lowest performing model with a training accuracy of 65.98%, produced a higher F1 score in the test process compared to the BERT model. The BERT model tended to produce a high level of false positives with a precision of 31.32% in the test process. Although it achieved an accuracy of 77.65% in the test process, it showed an unbalanced prediction tendency between classes with an F1 score of 27.15%.

The LSTM model achieved 76.01% accuracy in training and 63.96% accuracy in test. This model produced higher accuracy and F1 scores than the CNN model in both training and test phases.

In summary, the LLM's (GPT-2 and Llama3.1-8B), outperformed both transformer-based models (BERT, RoBERTa) and classical deep learning models (LSTM, CNN) in the customer complaint score classification problem.

TABLE VIII.
TRAIN METRICS OF MODELS DATASET 2

Model	Accuracy	Precision	Recall	F1-Score
Llama3.1-8B	0.9149	0. 9140	0. 9149	0.9139
gpt2	0.8999	0.8991	0.8999	0.8993
RoBERTa	0.9009	0.9003	0.9009	0.9005
BERT	0.7828	0.7847	0.7828	0.7716
LSTM	0.9504	0.9503	0.9504	0.9500
CNN	0.8941	0.8957	0.8941	0.8912

The experimental results for the second dataset are presented in Table VIII for the training data and Table IX for the test data. In general, LLMs performed better than deep

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TABLE IX.
TEST METRICS OF MODELS DATASET 2

Model	Accuracy	Precision	Recall	F1-Score
Llama3.1-8B	0.8999	0.8991	0.8999	0.8993
gpt2	0.8863	0.8846	0.8863	0.8849
RoBERTa	0.8877	0.8882	0.8877	0.8877
BERT	0.5614	0.3151	0.5614	0.4037
LSTM	0.8736	0.8722	0.8736	0.8722
CNN	0.8581	0.8589	0.8581	0.8534

learning-based models in classifying customer complaints. Among these models, Llama3.1-8B stood out with its lack of overfitting and high fit to the dataset. The Llama3.1-8B model achieved 91.49% accuracy and 91.39% F1 score in the training process, and 89.99% accuracy and 89.93% F1 score in the test process, showing the highest test performance among all models. The closeness between the training and test metrics shows that the model learns in a consistent and balanced manner. However, it should be noted that in return for this high performance, the Llama3.1-8B model requires significant GPU resources.

Although the LSTM model achieved the highest accuracy in the training process, it ranked fourth in the table in the test process. This indicates that the LSTM model is prone to overlearning on customer complaint data. LSTM has the most significant overfitting tendency among the models evaluated in the related table.

The BERT model, on the other hand, performed poorly in both training and test processes. This suggests that the model is not suitable for this problem with the current parameters. Possible reasons for this poor performance may include the fact that the BERT model does not include specific text types such as customer complaints in the pretraining process and that the hyperparameters were not set appropriately for the model in the experimental process.

Compared to BERT, the RoBERTa model performed better on the customer complaint classification task due to its dynamic masking strategy and pretraining process with a larger batch size. With these advantages, RoBERTa outperformed BERT in the classification task.

In the classification process performed with the CNN model, the 89.41% accuracy value obtained in training decreased to 85.81% in the test process. With an accuracy of 85.81% and an F1 score of 85.34%, CNN performed worse than the GPT-2 and RoBERTa models. In addition, the CNN model showed a similar tendency to overfitting as the LSTM.

Overall, the Llama3.1-8B model outperformed the deep learning architectures in the customer complaint classification problem. The BERT model showed the lowest performance in both training and test processes with 78.28% training accuracy and 56.14% test accuracy; it also showed the most significant overfitting tendency by showing the largest difference between these two processes. The BERT model also performed poorly in predicting unbalanced classes in the dataset with 31.51% precision, 56.14% recall and 40.37% F1 score in the test process.

As a result, the Llama3.1-8B model showed the highest and most consistent success in the customer complaint classification problem. Although GPT-2 and RoBERTa fall behind this model, they can be considered as worthy alternatives with their high classification performance. Especially when criteria such as model size and processing time are taken into consideration, these two models can offer usable solutions with lower resource consumption.

VI. CONCLUSION AND FUTURE WORK

This study aims to facilitate the early detection of customer issues by enterprises and organizations through the automatic classification of client complaints, consequently expediting resolution processes and enhancing customer satisfaction. This classification system facilitates the prioritization of complaints, enabling prompt resolution of issues necessitating urgent attention, thus optimizing the customer experience and service quality. In this regard, two distinct datasets, one comprising six classes and the other five classes, were evaluated utilizing standard deep learning techniques (LSTM, CNN), transformer-based architectures (BERT, Roberta), and LLMs (GPT-2, Llama3.1-8B). The results indicated that the models predominantly demonstrated superior classification performance on the second dataset. This condition can be ascribed to the dataset's more equitable class distribution. In the initial dataset, the GPT-2 model exhibited strong classification performance, but in the subsequent dataset, the Llama3.1-8B model achieved superior accuracy and F1 scores during both training and test phases. In the second dataset, deep learning models produced higher accuracy and f1 scores than LLMs in the train phase, but lagged behind LLMs in the test phase, indicating that although deep learning models are prone to overfitting, LLMs are more reliable.

Notwithstanding the diverse class distribution in the datasets, the significant performance exhibited by the LLMs suggests that these models possess robust generalization capabilities. GPT-2 has consistently performed across both datasets, indicating its suitability for customer complaint classification tasks. Conversely, the BERT model yielded the lowest F1 scores in the training and test phases, suggesting that its existing settings were inadequate for addressing this problem.

The findings suggest that AI-driven complaint classification systems possess significant applicability in customer-focused industries. In the telecommunications industry, the instantaneous categorization of consumer grievances facilitates the swift assessment of issues like network malfunctions or billing discrepancies, allowing prompt action in service interruptions. In the e-commerce and retail sectors, classifying complaints by product, shipping, or payment processes can enhance the efficiency of customer service administration. Applying these classifications in customer care centers or chatbot systems might streamline the allocation of pertinent complaints to the appropriate departments, thus minimizing processing durations. Furthermore, future communication infrastructures,

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termed 6G technologies, will leverage AI-assisted data analysis systems to yield substantial benefits in optimizing network performance and perpetually monitoring and increasing consumer experience. Integrating these systems with advanced security solutions such as Quantum Key Distribution (QKD) will add a new dimension to data privacy and security.

In this context, in today's world, where customer data is highly sensitive, it is of great importance that the artificial intelligence systems developed are not only cloud-based but also designed to operate on-premise (on local servers). Especially in high-security sectors such as finance, healthcare, public, and telecommunications, analyzing data without transferring it to an external environment is critical for compliance with legal regulations and protecting customer trust.

Future studies aim to make decision mechanisms transparent through hyperparameter optimization of models and Explainable AI approaches. Especially in regulated sectors, increasing the understandability of decisions is critical for model reliability. Moreover, developing lightweight and fast models that operate with lower resource consumption will facilitate the integration of these technologies into mobile platforms and real-time systems. In this context, it is possible to commercialize customer complaint classification systems on a broader scale and adapt them to different fields.

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